

IMSAXES

A PROJECT REPORT

Submitted by

ARPILKUMAR SANJAYBHAI KHUNT

220170107055

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Vishwakarma Government Engineering College

Near Visit circle, Sabarmati-Koba highway, Chandkheda, Ahmedabad,
Gujarat, India-382424

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **IMSAXES** has been carried out by **Khunt Arpilkumar Sanjaybhai (220170107055)** under my guidance in partial fulfilment for the degree of Bachelor of Engineering in Computer Engineering, 8th Semester of Gujarat Technological University, Ahmadabad during the academic year 2025-26.

Prof. Jaimin M. Shroff
Internal Guide

Prof. Kajal S. Patel
Head of the Department



Vishwakarma Government Engineering College Chandkheda, Ahmedabad

DECLARATION

I hereby declare that the Internship/ Project Report submitted along with the Internship entitled **IMSAXES** submitted in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering to Gujarat Technological University, Ahmadabad, is a Bonafide record of original project work carried out by me at **Meditab** under the supervision of **Nevil Patel** and that no part of this report has been directly copied from any student's reports or taken from any other source, without providing due reference.

Name of Student

ARPILKUMAR SANJAYBHAI KHUNT

Sign Student

ACKNOWLEDGEMENT

I would like to take this opportunity to extend my heartfelt appreciation to everyone who played a role in the successful completion of my internship project.

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Thank you,

ARPILKUMAR SANJAYBHAI KHUNT

220170107055

ABSTRACT

Purpose of Internship:

The internship provided a valuable opportunity to develop real-world technical and soft skills through hands-on project work. It helped in strengthening problem-solving capabilities, communication, and collaborative skills while gaining deep insights into test automation frameworks and practices.

Scope of Internship:

*The scope included developing robust and scalable automation frameworks using tools such as **Selenium**, **TestNG**, **Maven**, **Extent Reports**, and **UFT**. The focus was on creating maintainable and reusable scripts capable of handling a wide range of testing scenarios, including both sanity and regression testing.*

Work Done:

The internship involved two primary projects:

1. IMSAXES:

Worked on the development of automation scripts for a company-internal platform named "AXES" (Automation Execution Environment for Selenium and IMS). The system automated sanity and IMS testing of internal web applications using test suites provided by the QA team. The scripts were modular and designed to ensure minimal human intervention during execution.

2. Smart Support System (Quickcap Support Bot):

Smart Support System (Quickcap Support Bot) is an AI-powered application built using a Retrieval-Augmented Generation (RAG) approach to deliver accurate and context-aware responses. The system consists of two main modules: an ingestion pipeline that processes PDFs and other resources to build a dynamic knowledge base, and a conversational interface for real-time user interaction. It leverages streaming capabilities via Streamlit to provide responses word-by-word, enhancing user experience with minimal latency.

Together, both projects contributed to a significant reduction in manual testing efforts and knowledge retrieval, increased test coverage, and enhanced the maintainability of test suites - thereby improving overall product quality and testing efficiency.

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Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
SQL	Structured Query Language
IMS	Intelligent Medical Software
AXES	Automation Export Environmental Software
EHR	Electronic Health Record
IDE	Intelligent Development Environment
JDK	Java Development Kit
QA	Quality Assurance

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